1. **Schema Validation (Star Schema)**
   **Objective:** Verify that all star schema tables and constraints were created correctly.

   **SQL Commands used:**

```
SHOW TABLES;

DESCRIBE dim_student;
DESCRIBE dim_course;
DESCRIBE dim_instructor;
DESCRIBE dim_date;
DESCRIBE fact_enrollments;
```

   **Screenshots:**

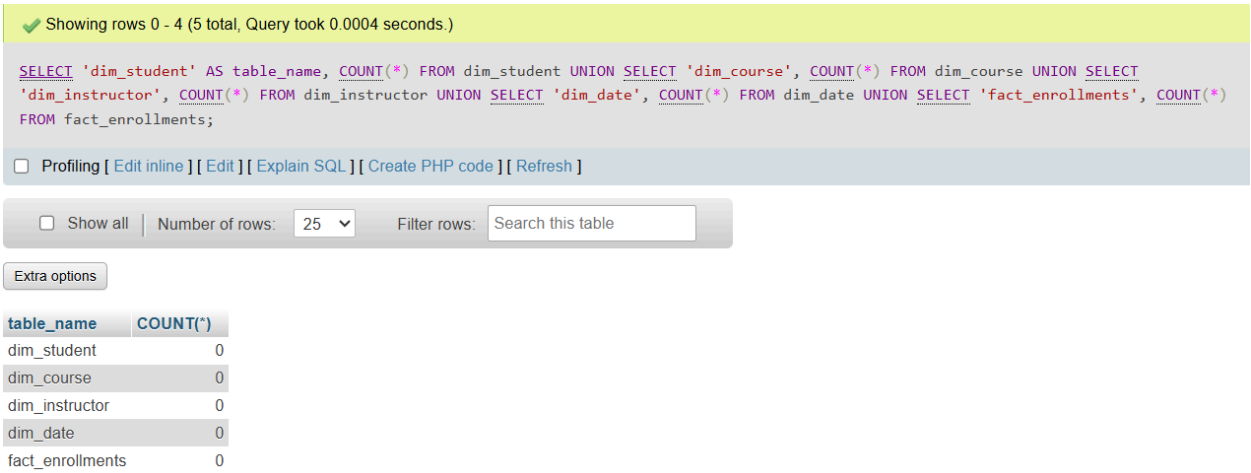
   **Result:** All star schema tables were successfully created with correct columns, data types, primary keys, and foreign key constraints.

2. **Data Population (Star Schema Validation)**
   **Objective:** Ensure that all star schema tables contain sufficient data for analysis.

   **SQL Commands used:**

```
SELECT 'dim_student' AS table_name, COUNT(*) FROM dim_student
UNION
SELECT 'dim_course', COUNT(*) FROM dim_course
UNION
SELECT 'dim_instructor', COUNT(*) FROM dim_instructor
UNION
SELECT 'dim_date', COUNT(*) FROM dim_date
UNION
SELECT 'fact_enrollments', COUNT(*) FROM fact_enrollments;
```

   **Screenshots:**




| table_name | COUNT(*) |
|---|---|
| dim_student | 0 |
| dim_course | 0 |
| dim_instructor | 0 |
| dim_date | 0 |
| fact_enrollments | 0 |

   **Result:** The query confirms that all star schema tables contain data. The dataset is complete and sufficient for business intelligence analysis.

3. **Data Integrity Validation**
   **Objective:** Ensure that all fact records correctly reference dimension tables and contain no orphan records.

## SQL Commands used:

```
SELECT *
FROM fact_enrollments f
LEFT JOIN dim_student s ON f.studentID = s.studentID
WHERE s.studentID IS NULL;

SELECT *
FROM fact_enrollments f
LEFT JOIN dim_course c ON f.courseID = c.courseID
WHERE c.courseID IS NULL;

SELECT *
FROM fact_enrollments f
LEFT JOIN dim_instructor i ON f.instructorID = i.instructorID
WHERE i.instructorID IS NULL;

SELECT *
FROM fact_enrollments f
LEFT JOIN dim_date d ON f.dateID = d.dateID
WHERE d.dateID IS NULL;
```

## Screenshots:

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0009 seconds.)

```
SELECT * FROM fact_enrollments f LEFT JOIN dim_student s ON f.studentID = s.studentID WHERE s.studentID IS NULL;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

| enrollmentID | studentID | courseID | instructorID | dateID | grade | studentID | firstName | lastName | email |
|---|---|---|---|---|---|---|---|---|---|

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
SELECT * FROM fact_enrollments f LEFT JOIN dim_course c ON f.courseID = c.courseID WHERE c.courseID IS NULL;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

| enrollmentID | studentID | courseID | instructorID | dateID | grade | courseID | courseName | instructorID |
|---|---|---|---|---|---|---|---|---|

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
SELECT * FROM fact_enrollments f LEFT JOIN dim_instructor i ON f.instructorID = i.instructorID WHERE i.instructorID IS NULL;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

| enrollmentID | studentID | courseID | instructorID | dateID | grade | instructorID | name | email |
|---|---|---|---|---|---|---|---|---|

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
SELECT * FROM fact_enrollments f LEFT JOIN dim_date d ON f.dateID = d.dateID WHERE d.dateID IS NULL;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

| enrollmentID | studentID | courseID | instructorID | dateID | grade | dateID | fullDate | year | month | day |
|---|---|---|---|---|---|---|---|---|---|---|

**Result:** All queries returned zero rows, confirming that there are no orphan records. Referential integrity is properly maintained across the star schema.

4. **Relationship (JOIN) Validation**
**Objective:** Verify that relationships between fact and dimension tables return correct and meaningful data.

**SQL Commands used:**

```
SELECT
    f.enrollmentID,
    s.firstName,
    c.courseName,
    i.name AS instructor_name,
    d.fullDate
FROM fact_enrollments f
JOIN dim_student s ON f.studentID = s.studentID
JOIN dim_course c ON f.courseID = c.courseID
JOIN dim_instructor i ON f.instructorID = i.instructorID
JOIN dim_date d ON f.dateID = d.dateID
LIMIT 10;
```

**Screenshots:**

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0003 seconds.)

```
SELECT f.enrollmentID, s.firstName, c.courseName, i.name AS instructor_name, d.fullDate FROM fact_enrollments f JOIN dim_student s ON
f.studentID = s.studentID JOIN dim_course c ON f.courseID = c.courseID JOIN dim_instructor i ON f.instructorID = i.instructorID JOIN
dim_date d ON f.dateID = d.dateID LIMIT 10;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

| enrollmentID | firstName | courseName | instructor_name | fullDate |
|---|---|---|---|---|

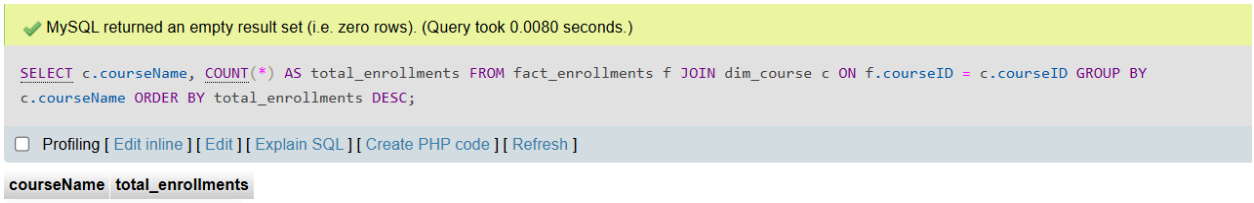
**Result:** The query successfully retrieved related data from all dimension tables, confirming that the star schema relationships are correctly implemented and functional.

5. **BI Query Validation**
   **Objective:** Verify that the star schema supports business intelligence queries and aggregations.

   **SQL Commands used:**

```
SELECT
    c.courseName,
    COUNT(*) AS total_enrollments
FROM fact_enrollments f
JOIN dim_course c ON f.courseID = c.courseID
GROUP BY c.courseName
ORDER BY total_enrollments DESC;
```

   **Screenshots:**




| courseName | total_enrollments |
|---|---|

   **Result:** The query successfully generated aggregated results, confirming that the star schema supports business intelligence analysis such as grouping and summarization.

6. **Time-Based Analysis Validation**
   **Objective:** Verify that the date dimension supports time-based analytics.

   **SQL Commands used:**

```
SELECT
    d.year,
    d.month,
    COUNT(*) AS total_enrollments
FROM fact_enrollments f
JOIN dim_date d ON f.dateID = d.dateID
GROUP BY d.year, d.month
ORDER BY d.year, d.month;
```

## Screenshots:

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0008 seconds.)

```
SELECT d.year, d.month, COUNT(*) AS total_enrollments FROM fact_enrollments f JOIN dim_date d ON f.dateID = d.dateID GROUP BY d.year,
d.month ORDER BY d.year, d.month;
```

Profiling [ Edit inline ] [ Edit ] [ Explain SQL ] [ Create PHP code ] [ Refresh ]

| year | month | total_enrollments |
|---|---|---|

**Result:** The query successfully generated time-based analytical results, confirming that the star schema supports trend analysis and reporting over time.